MIL-C-24643/21B
3 October 1986
SUPERSEDING
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8 November 1985
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28 September 1984

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPES LSTCJU AND LSTCTU (INCLUDING VARIATIONS LSTCJA AND LSTCTA)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the cable described herein shall consist of this specification and the latest issue of MIL-C-24643.

REQUIREMENTS:

Qualification required.

Construction (watertight with circuit integrity)

First - One iron and one constantan conductor in type LSTCJU, for type J thermocouple in accordance with ANSI MC96.1; one copper and one constantan in type LSTCTU, for type T thermocouple in accordance with ANSI MC96.1; (see table I for size). All conductors uncoated.
Second - Extruded silicone rubber insulation (see table I for diameter).
Third - Glass braid.
Fourth - Braid covering. Printed identification as follows: Type LSTCJU shall have the legend "8 (GRAY)" printed on the iron conductor, and the legend "3 (RED)" printed on the constantan conductor. Type LSTCTU shall have the legend "6 (BLUE)" printed on the copper conductor, and the legend "3 (RED)" printed on the constantan conductor (see table I for diameter).
Fifth - The two conductors shall be cabled together with a lay of 16 to 24 times the pitch diameter. Fillers shall be used to form a firm, well-rounded assembly.
Sixth - Binder or combination binder and barrier.

(B) denotes changes.

AMSC N/A FSC 6145
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Construction variant for types LSTCJU and LSTCTU

Seventh - Cross-linked polyolefin jacket. Nominal thickness 0.050 inch. Cable surface marking required.

Construction variant for types LSTCTA and LSTCJA

(B) Seventh - Cross-linked polyolefin jacket. Nominal thickness 0.050 inch. Cable surface marking not required.
Eighth - Braided metal armor and paint.

TABLE I. Details.

(B) Military part no. M24643/21	Type and size	Number of strands and strand diameter (inch)	Diameter over insulation minimum (inch)	Diameter over braid covering maximum (inch)	Overall diameter minimum (inch)	Overall diameter maximum (inch)	Cold bending mandrel diameter maximum (inches)
-01UN	LSTCJU-4	7/.025	0.105	0.140	0.400	0.430	6
-02UN	LSTCTU-4	7/.025	.105	.140	.400	.430	6
(B) -03AN	LSTCJA-4	7/.025	.105	.140	---	.480	6
(B) -04AN	LSTCTA-4	7/.025	.105	.140	---	.480	6

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Voltage withstand - volts, root mean square, minimum....	3000
Insulation resistance - megohms/1000 feet, minimum.....	500
ANSI MC96.1 calibration limits of error:	
(a) Types LSTCJA and LSTCJU.....	± 2.2°C or ± 0.75 percent
(b) Types LSTCTA and LSTCTU.....	± 1°C or ± 0.75 percent
Conductor continuity.....	No failure
Group A:	
Visual and dimensional.....	No failure
Watertightness - see MIL-C-24643 for limits of water leakage.....	No failure
Abrasion resistance - scrapes, minimum.....	250
Crack resistance.....	No damage

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EXAMINATION AND TESTS: (Continued)

	Requirements
Group B:	
Gas flame - 1 hour	No failure
Drip - 95 ± 1°C	Zero
Tear - pounds per inch thickness, minimum (ASTM D 470)	35
Cold bending, cable - (see table I for mandrel diameter)	No damage
Physicals (unaged)	
Insulation (extruded)	
Tensile strength - lbf/in^2, minimum	750
Elongation - percent, minimum	125
Jacket (cable)	
Tensile strength - lbf/in^2, minimum	1300
Elongation - percent, minimum	160
Group C:	
Physicals (aged)	
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged, minimum	60
Hot oil immersion	
Tensile strength - lb/in^2, minimum	50
Elongation - percent, minimum	50
Shrinkage	No failure
Heat distortion - percent of unaged, maximum	30
Permanence of printing (conductor) - cycles, minimum	50
Permanence of printing (jacket) - cycles, minimum	250
Cable filler removability	No failure
Armor (types LSTCTA and LSTCJA only) - conformance to material, construction and coverage	No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B and C, plus the following:

Cold working (minus 20 ± 2°C)	No damage
Gas flame (3 hours)	No failure
Specific gravity of extruded silicone insulation, maximum	1.55
Aging and compatibility (cable) (125 ± 5°C)	No failure
(B) Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2

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QUALIFICATION INSPECTION: (Continued)

	Requirements
Flame propagation (cable)	No failure
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	—
(B) Insulation	35
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
LSTCJU-4 and LSTCTU-4	2000

NOTE: Not for Air Force use.

Custodians:
Army - MI
Navy - SH

Review activities:
Army - AV, CR, ER
Navy - EC
DLA - IS

User activities:
Army - ME, AR, AL
Navy - CG

Preparing activity:
Navy - SH
(Project 6145-1005-21)

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